ECE 4502/6502 & CS 6501: Machine Learning on Graphs

Lecture 3. Network Measures (I)

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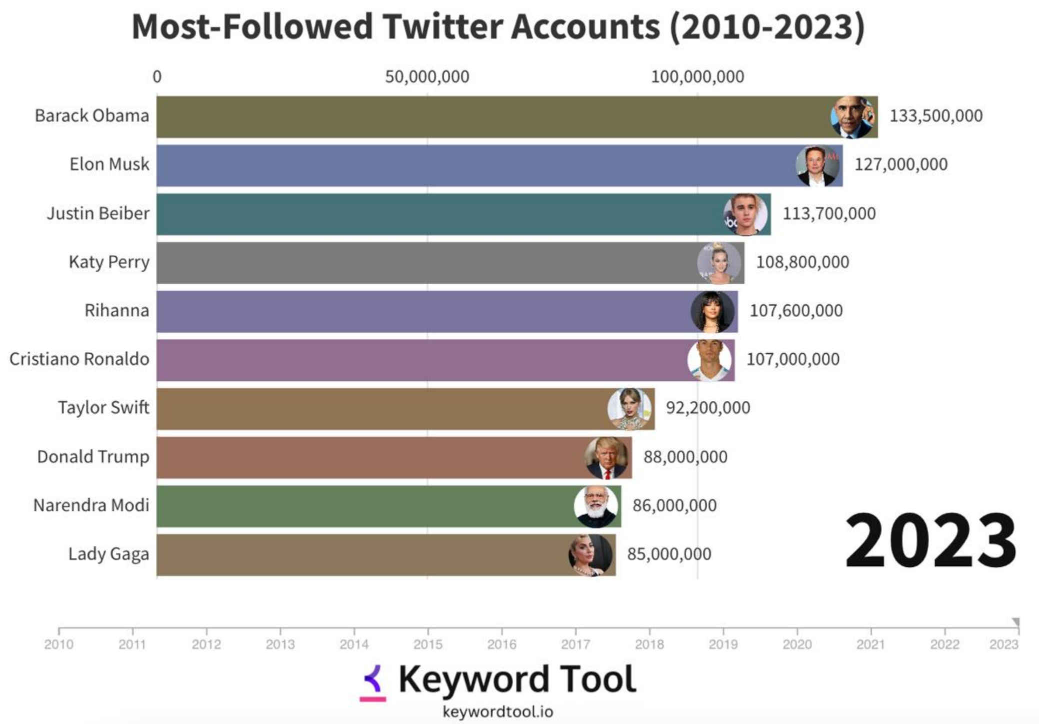
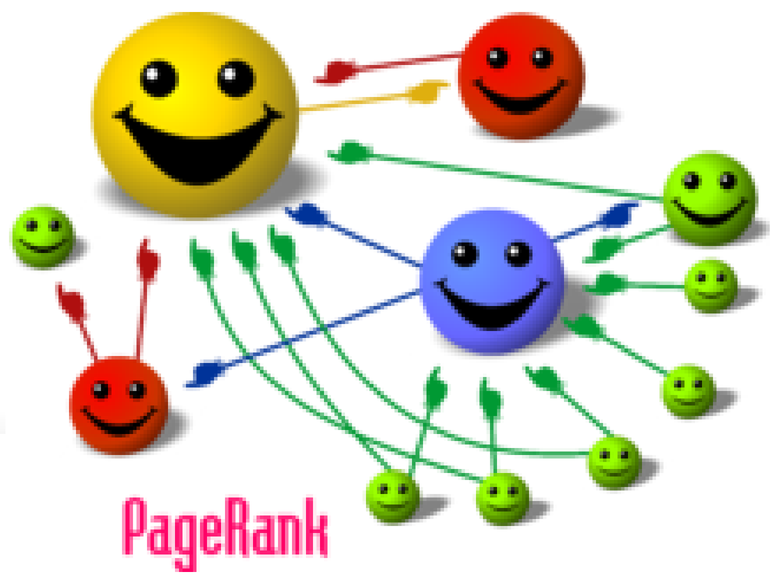
Spring 2025, University of Virginia

What to measure?





Why Do We Need Measures?

- Who are the most important nodes in the network?
    - **Centrality**
- How to characterize the interaction patterns of edges?
    - **Reciprocity and Transitivity**
    - **Balance and Status**
- How to measure the similarity between nodes?
    - **Similarity measures**
- To answer these and similar questions, one first needs to define measures for quantifying **centrality**, **level of interactions**, and **similarity**, among others.

Centrality

Centrality defines how important a node is within a network

Centrality in terms of those who you are connected to

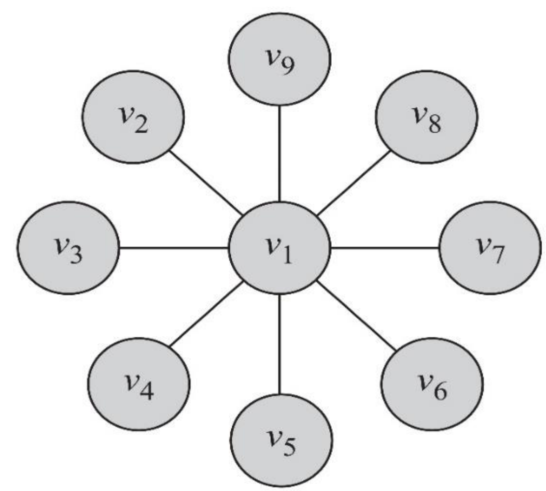
Degree Centrality

- **Degree centrality**: ranks nodes with more connections higher in terms of centrality

$$C_d(v_i) = d_i$$

- d_i is the degree (number of friends) for node v_i
  - i.e., the number of length-1 paths (can be generalized)

In this graph, degree centrality for node v_1 is d_1=8 and for all others is $d_j = 1, j \neq 1$



Degree Centrality in Directed Graphs

- In directed graphs, we can either use the in-degree, the out-degree, or the combination as the degree centrality value:
- In practice, mostly in-degree is used

$$
\begin{aligned}
C_d(v_i) &= d_i^{\text{in}} && \textit{(prestige)} \\
C_d(v_i) &= d_i^{\text{out}} && \textit{(gregariousness)} \\
C_d(v_i) &= d_i^{\text{in}} + d_i^{\text{out}}
\end{aligned}
$$

d_i^{out} is the number of outgoing links for node v_i

Normalized Degree Centrality

Comparability of degree centrality across platforms

Users: 1 billion

Users: 335.7 million

- Normalized by the maximum possible degree

$$C_d^{\text{norm}}(v_i) = \frac{d_i}{n-1}$$

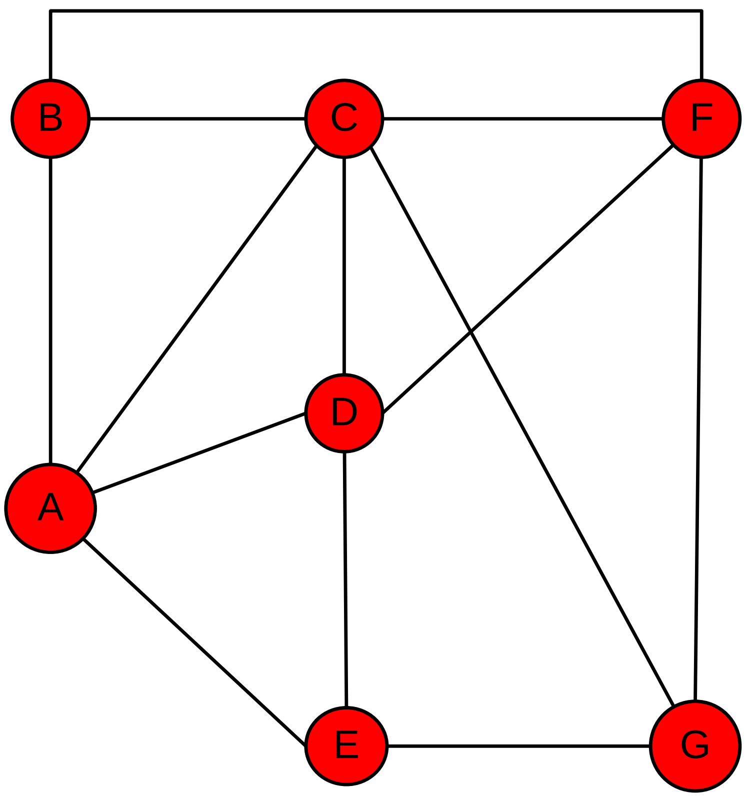
- Normalized by the maximum degree

$$C_d^{\max}(v_i) = \frac{d_i}{\max_j d_j}$$

- Normalized by the degree sum

$$C_d^{\text{sum}}(v_i) = \frac{d_i}{\sum_j d_j} = \frac{d_i}{2|E|} = \frac{d_i}{2m}$$

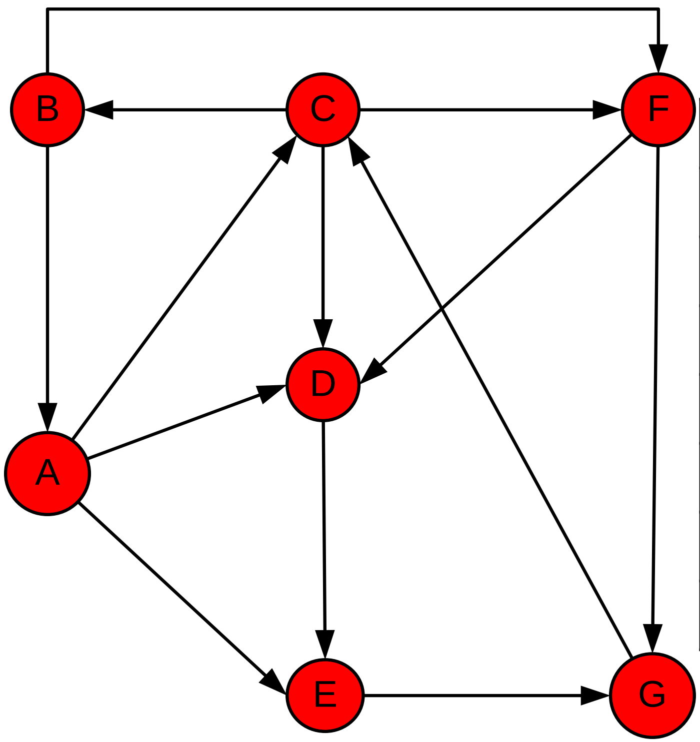
Degree Centrality (undirected Graph) Example



Node	Degree	Centrality	Rank
A	4	2/3	2
B	3	1/2	5
C	5	5/6	1
D	4	2/3	2
E	3	1/2	5
F	4	2/3	2
G	3	1/2	5

Normalize by the maximum <u>possible</u> degree

Degree Centrality (Directed Graph) Example



Node	In-Degree	Out-Degree	Centrality	Rank
A	1	3	1/2	1
B	1	2	1/3	3
C	2	3	1/2	1
D	3	1	1/6	5
E	2	1	1/6	5
F	2	2	1/3	3
G	2	1	1/6	5

Normalized by the maximum possible degree

$$C_d^{\text{norm}}(v_i) = \frac{d_i}{n-1}$$

Eigenvector Centrality

- Problem of degree centrality
  - High degree centrality ≠ high importance

- Solution: Having more **important friends** provides a stronger signal
- Eigenvector centrality generalizes degree centrality by incorporating the importance of the neighbors (undirected)
- For directed graphs, we can use incoming neighbors or outgoing neighbors

Formulation

- Let's assume the eigenvector centrality of a node is $c_e(v_i)$ (unknown)

- We would like $c_e(v_i)$ to be higher when important neighbors (node v_j with higher $c_e(v_j)$) point to us
  - Incoming or outgoing neighbors?
  - For incoming neighbors $A_{j,i} = 1$

Formulation Cont.

- We can assume that v_i's centrality is obtained by the summation of its neighbors' centralities

$$c_e(v_i) = \sum_{j=1}^{n} A_{j,i} c_e(v_j)$$

- Is this summation bounded?

$$c_e(v_i) = \frac{1}{\lambda} \sum_{j=1}^{n} A_{j,i} c_e(v_j)$$

We have to normalize!

λ: some fixed constant

Eigenvector Centrality (Matrix Formulation)

- Let $\mathbf{C}_e = (C_e(v_1), C_e(v_2), \ldots, C_e(v_n))^T$

$$\rightarrow \lambda \mathbf{C}_e = A^T \mathbf{C}_e$$

- This means that $\mathbf{C}_e$ is an eigenvector of adjacency matrix A^T (or A when undirected) and λ is the corresponding eigenvalue

- Which eigenvalue-eigenvector pair should we choose?

Eigenvector Centrality Cont.

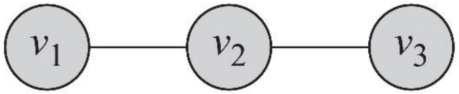
- Which eigenpair should we choose?
  - Desired property: positive eigenvector values
- Perron-Frobenius Theorem
  - For positive matrix or irreducible non-negative matrix ***A*** (always holds for undirected graphs)
  - The eigenvalue $\mathbf{C_e}$ associated with λ_{max} is strictly positive

Theorem 1 (Perron-Frobenius Theorem). *Let $A \in \mathbb{R}^{n \times n}$ represent the adjacency matrix for a [strongly] connected graph or $A : A_{i,j} > 0$ (i.e. a positive n by n matrix). There exists a positive real number (Perron-Frobenius eigenvalue) $\lambda_{\max}$, such that $\lambda_{\max}$ is an eigenvalue of A and any other eigenvalue of A is strictly smaller than $\lambda_{\max}$. Furthermore, there exists a corresponding eigenvector $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ of A with eigenvalue $\lambda_{\max}$ such that $\forall v_i > 0$.*

Eigenvector Centrality

- Key steps
  - Perform eigen-decomposition of adjacency matrix A
  - Select the largest eigenvalue λ and the corresponding eigenvector $\mathbf{c_e}$
  - Based on the Perron-Frobenius theorem, all the components of $\mathbf{c_e}$ will be positive
  - The components of $\mathbf{c_e}$ are the eigenvector centralities for the graph.

Eigenvector Centrality: Example 1



$$\lambda \mathbf{C}_e = A\mathbf{C}_e \qquad (A - \lambda I)\mathbf{C}_e = 0 \quad \mathbf{C}_e = [u_1 \ u_2 \ u_3]^T$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0-\lambda & 1 & 0 \\ 1 & 0-\lambda & 1 \\ 0 & 1 & 0-\lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$det(A - \lambda I) = \begin{vmatrix} 0-\lambda & 1 & 0 \\ 1 & 0-\lambda & 1 \\ 0 & 1 & 0-\lambda \end{vmatrix} = 0$$

$$(-\lambda)(\lambda^2 - 1) - 1(-\lambda) = 2\lambda - \lambda^3 = \lambda(2 - \lambda^2) = 0$$

Eigenvalues are

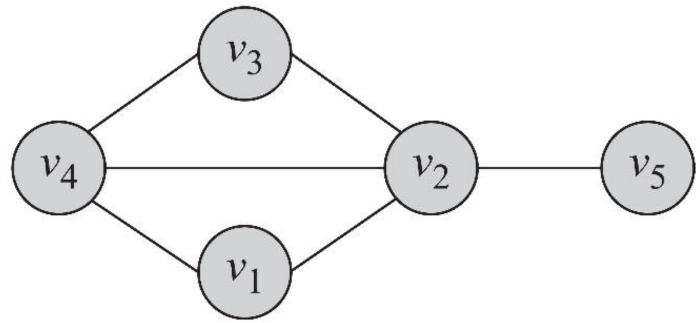
$$(-\sqrt{2}, 0, \boxed{+\sqrt{2}})$$

Largest Eigenvalue

Corresponding eigenvector (assuming $\mathbf{C}_e$ has norm 1)

$$\begin{bmatrix} 0-\sqrt{2} & 1 & 0 \\ 1 & 0-\sqrt{2} & 1 \\ 0 & 1 & 0-\sqrt{2} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad \mathbf{C}_e = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 1/2 \\ \sqrt{2}/2 \\ 1/2 \end{bmatrix}$$

Eigenvector Centrality: Example 2



$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \boldsymbol{\lambda} = (2.68, -1.74, -1.27, 0.33, 0.00)$$

Eigenvalues Vector

$$\lambda_{max} = 2.68 \Rightarrow C_e = \begin{bmatrix} 0.4119 \\ 0.5825 \\ 0.4119 \\ 0.5237 \\ 0.2169 \end{bmatrix}$$

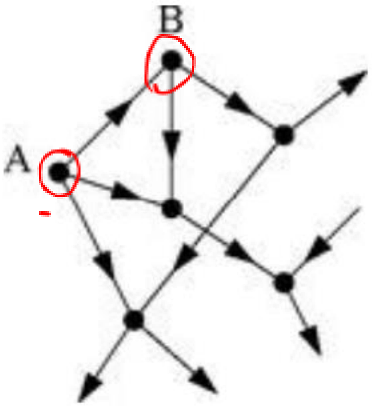
Katz Centrality

Elihu Katz

- A major problem with eigenvector centrality arises when it deals with directed graphs

$$c_e(v_i) = \frac{1}{\lambda} \sum_{j=1}^{n} A_{j,i} c_e(v_j)$$

- Centrality only passes over *outgoing* edges and in special cases such as when a node is in a directed acyclic graph centrality becomes zero



Katz Centrality Cont.

- To resolve this problem we add bias term β to the centrality values for all nodes

Eigenvector Centrality

$$C_{\mathrm{Katz}}(v_i) = \boxed{\alpha \sum_{j=1}^{\mathrm{n}} A_{j,i} C_{\mathrm{Katz}}(v_j)} + \beta$$

$$c_e(v_i) = \frac{1}{\lambda} \sum_{j=1}^{n} A_{j,i} c_e(v_j)$$

Katz Centrality Cont.

$$C_{\mathrm{Katz}}(v_i) = \alpha \sum_{j=1}^{\mathrm{n}} A_{j,i} C_{\mathrm{Katz}}(v_j) + \beta$$

Controlling term

Bias term

Rewriting equation in a vector form

$$\mathbf{C}_{\mathrm{Katz}} = \alpha A^T \mathbf{C}_{\mathrm{Katz}} + \beta \mathbf{1}$$

vector of all 1's

Katz centrality: $\mathbf{C}_{\mathrm{Katz}} = \beta(\underline{\mathbf{I} - \alpha A^T})^{-1} \cdot \mathbf{1}$

$\alpha \neq \frac{1}{\lambda}$

Katz Centrality Cont.

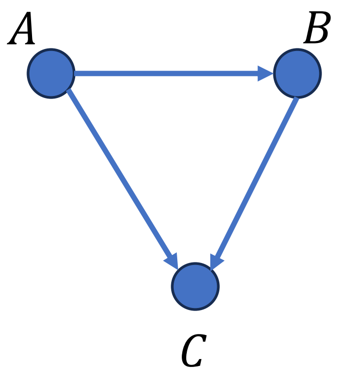
$$C_{\text{Katz}}(v_i) = \alpha \sum_{j=1}^{\text{n}} A_{j,i} C_{\text{Katz}}(v_j) + \beta$$

- When α=0, the eigenvector centrality is removed and all nodes get the same centrality value β
  - As α gets larger the effect of β is reduced
- For the matrix $(I - \alpha A^T)$ to be invertible, we must have
  - $det(I - \alpha A^T) \neq 0$
  - By rearranging we get $det(\mathrm{A}^{\mathrm{T}} - \alpha^{-1} I) = 0$
  - This is basically the characteristic equation
  - The characteristic equation **first** becomes zero when the largest eigenvalue equals α^{-1}

$\boxed{0 < \alpha < 1/\lambda_{max}}$

In practice we select $\boldsymbol{\alpha < 1/\lambda}$, where λ is the largest eigenvalue of $\boldsymbol{A^T}$

Katz Centrality Example 1



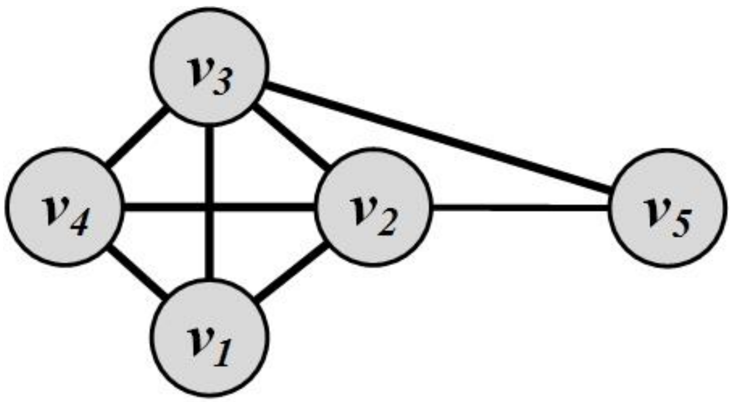
$$A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad A^T = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

- The largest eigenvalue for A^T is 1
- We assume α=0.25 and $\beta = 0.2$

$$\mathbf{C}_{\text{Katz}} = \beta(\mathbf{I} - \alpha A^T)^{-1} \cdot \mathbf{1} = \begin{bmatrix} \boxed{0.2} \\ 0.25 \\ 0.417 \end{bmatrix}$$

Not 0 anymore!

Katz Centrality Example 2



$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} = A^T$$

- The Eigenvalues are -1.68, -1.0, -1.0, 0.35, $\boxed{3.32}$
- We assume α=0.25 < $\boxed{1/3.32}$ and $\beta = 0.2$

$$\mathbf{C}_{Katz} = \beta(\mathbf{I} - \alpha A^T)^{-1} \cdot \mathbf{1} = \begin{bmatrix} 1.14 \\ 1.31 \\ 1.31 \\ 1.14 \\ 0.85 \end{bmatrix}$$

Most important nodes!

PageRank

- Problem with Katz Centrality:
  - In directed graphs, once a node becomes an authority (high centrality), it passes **all** its centrality along **all** of its out-links
- This is less desirable since not everyone known by a well-known person is well-known
- **Solution?**
  - We can divide the value of passed centrality by the number of outgoing links, i.e., out-degree of node
  - Each connected neighbor gets a fraction of the source node's centrality

PageRank Cont.

$$C_p(v_i) = \alpha \sum_{j=1}^{n} A_{j,i} \frac{C_p(v_j)}{d_j^{\text{out}}} + \beta$$

What if the degree is zero?

$$\begin{cases} d_j^{out} > 0 \\ D = diag(d_1^{\text{out}}, d_2^{\text{out}}, \ldots, d_n^{\text{out}}) \end{cases}$$

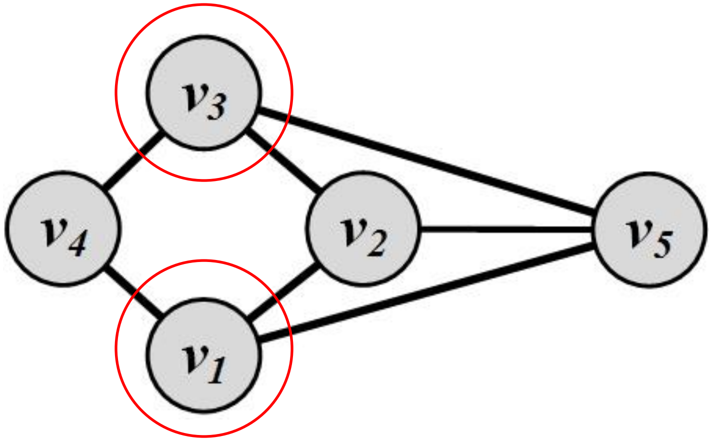
$$\mathbf{C}_p = \alpha A^T D^{-1} \mathbf{C}_p + \beta \mathbf{1}$$

$$\mathbf{C}_p = \beta (\mathbf{I} - \alpha A^T D^{-1})^{-1} \cdot \mathbf{1}$$

Similar to Katz Centrality, in practice, $\alpha < 1/\lambda$, where λ is the largest eigenvalue of $A^T D^{-1}$. In undirected graphs, the largest eigenvalue of $A^T D^{-1}$ is $\lambda = 1$; therefore, $\alpha < 1$.

PageRank Example

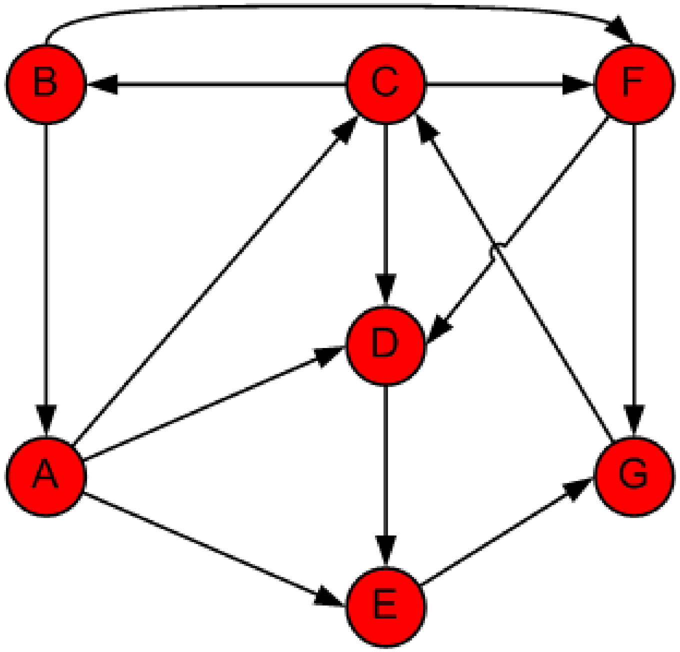
- We assume α=0.95 < 1 and and $\beta = 0.1$



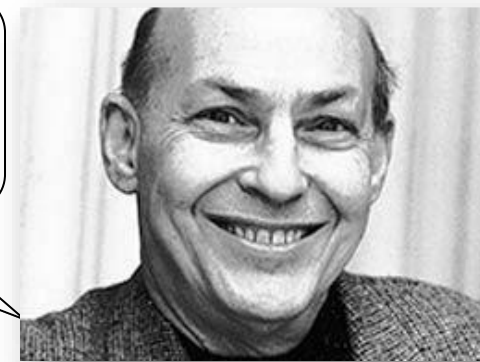
$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{C}_p = \beta(\mathbf{I} - \alpha A^T D^{-1})^{-1} \cdot \mathbf{1} = \begin{bmatrix} 2.14 \\ 2.13 \\ 2.14 \\ 1.45 \\ 2.13 \end{bmatrix}$$

PageRank Example – Alternative Approach



"You don't understand anything until you learn it more than one way"



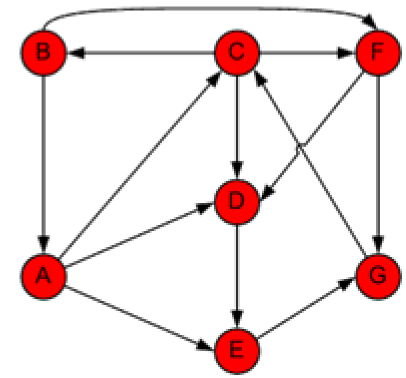
Marvin Minsky (1927-2016)

Using Power Method $O(n^2)$

α=1 and $\beta = 0$?

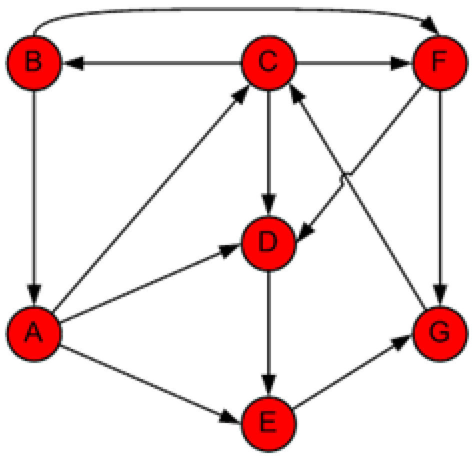
Step	A	B	C	D	E	F	G
0	1/7	1/7	1/7	1/7	1/7	1/7	1/7
1	B/2	C/3	A/3 + G	A/3 + C/3 + F/2	A/3 + D	C/3 + B/2	F/2 + E
	0.071	0.048	0.190	0.167	0.190	0.119	0.214

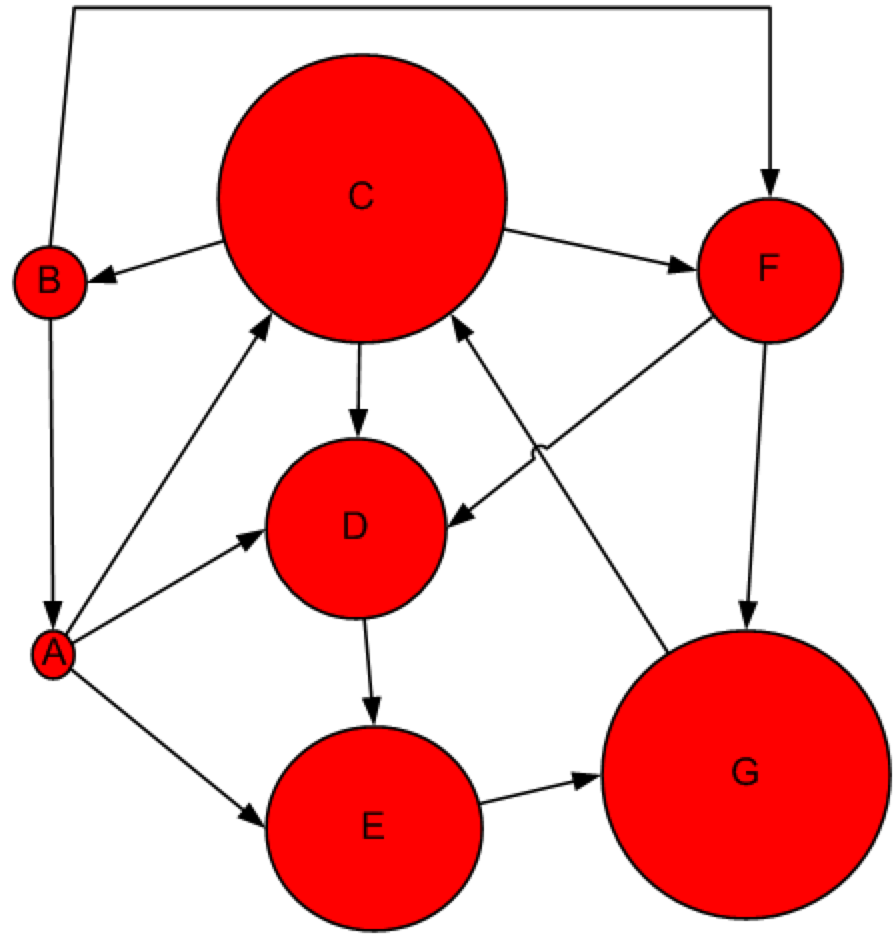
PageRank: Example



Step	A	B	C	D	E	F	G	Sum
1	0.143	0.143	0.143	0.143	0.143	0.143	0.143	1.000
2	0.071	0.048	0.190	0.167	0.190	0.119	0.214	1.000
3	0.024	0.063	0.238	0.147	0.190	0.087	0.250	1.000
4	0.032	0.079	0.258	0.131	0.155	0.111	0.234	1.000
5	0.040	0.086	0.245	0.152	0.142	0.126	0.210	1.000
6	0.043	0.082	0.224	0.158	0.165	0.125	0.204	1.000
7	0.041	0.075	0.219	0.151	0.172	0.115	0.228	1.000
8	0.037	0.073	0.241	0.144	0.165	0.110	0.230	1.000
9	0.036	0.080	0.242	0.148	0.157	0.117	0.220	1.000
10	0.040	0.081	0.232	0.151	0.160	0.121	0.215	1.000
11	0.040	0.077	0.228	0.151	0.165	0.118	0.220	1.000
12	0.039	0.076	0.234	0.148	0.165	0.115	0.223	1.000
13	0.038	0.078	0.236	0.148	0.161	0.116	0.222	1.000
14	0.039	0.079	0.235	0.149	0.161	0.118	0.219	1.000
15	0.039	0.078	0.232	0.150	0.162	0.118	0.220	1.000
Rank	7	6	1	4	3	5	2	

Effect of PageRank



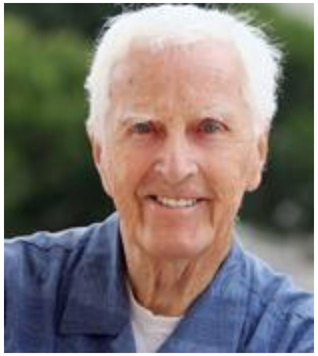


Node	Rank
A	7
B	6
C	1
D	4
E	3
F	5
G	2

Centrality in terms of how you connect others
(information broker)

Betweenness Centrality

Another way of looking at centrality is by considering how important nodes are in connecting other nodes



Linton Freeman

$$C_b(v_i) = \sum_{s \neq t \neq v_i} \boxed{\frac{\sigma_{st}(v_i)}{\sigma_{st}}} = 1$$

σ_{st} The number of shortest paths from vertex s to t – *a.k.a.* information pathways

$\sigma_{st}(v_i)$ The number of **shortest paths** from s to t that pass through v_i

Normalizing Betweenness Centrality

- In the best case, node v_i is on all shortest paths from s to t, hence, $\frac{\sigma_{st}(v_i)}{\sigma_{st}} = 1$

n

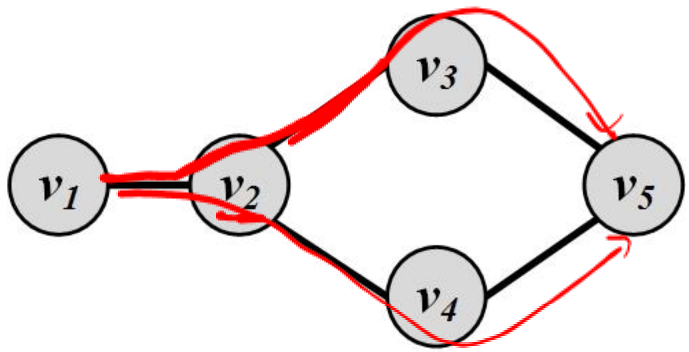
$2\binom{n-1}{2}$

$$C_b(v_i) = \sum_{s \neq t \neq v_i} \frac{\sigma_{st}(v_i)}{\sigma_{st}}$$
$$= \sum_{s \neq t \neq v_i} 1 = 2\binom{n-1}{2} = (n-1)(n-2)$$

Therefore, the maximum value is $(n-1)(n-2)$

Betweenness centrality: $C_b^{\text{norm}}(v_i) = \frac{C_b(v_i)}{2\binom{n-1}{2}}$

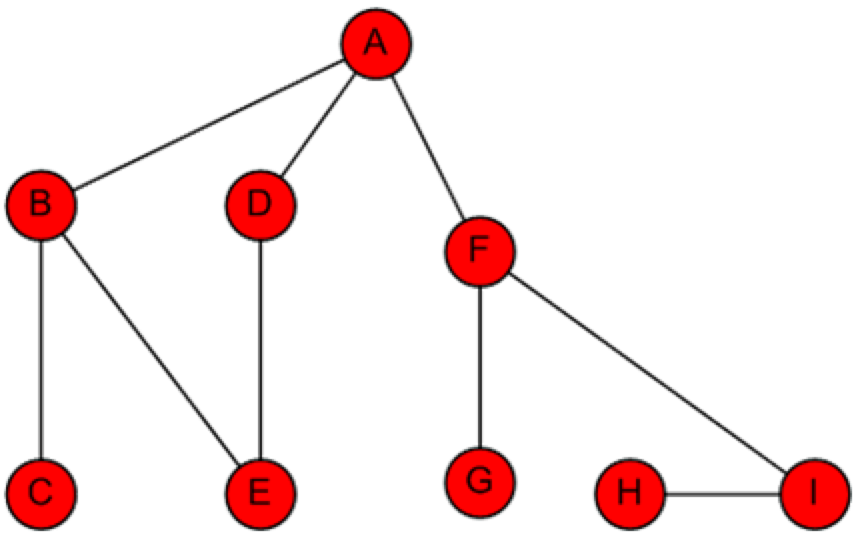
Betweenness Centrality: Example



$\frac{0}{1}$

$$
\begin{aligned}
C_b(v_2) &= 2 \times (\underbrace{(1/1)}_{s=v_1,t=v_3} + \underbrace{(1/1)}_{s=v_1,t=v_4} + \underbrace{(2/2)}_{s=v_1,t=v_5} + \underbrace{(1/2)}_{s=v_3,t=v_4} + \underbrace{0}_{s=v_3,t=v_5} + \underbrace{0}_{s=v_4,t=v_5}) \\
&= 2 \times 3.5 = 7, \\
C_b(v_3) &= 2 \times (\underbrace{0}_{s=v_1,t=v_2} + \underbrace{0}_{s=v_1,t=v_4} + \underbrace{(1/2)}_{s=v_1,t=v_5} + \underbrace{0}_{s=v_2,t=v_4} + \underbrace{(1/2)}_{s=v_2,t=v_5} + \underbrace{0}_{s=v_4,t=v_5}) \\
&= 2 \times 1.0 = 2, \\
C_b(v_4) &= C_b(v_3) = 2 \times 1.0 = 2, \\
C_b(v_5) &= 2 \times (\underbrace{0}_{s=v_1,t=v_2} + \underbrace{0}_{s=v_1,t=v_3} + \underbrace{0}_{s=v_1,t=v_4} + \underbrace{0}_{s=v_2,t=v_3} + \underbrace{0}_{s=v_2,t=v_4} + \underbrace{(1/2)}_{s=v_3,t=v_4}) \\
&= 2 \times 0.5 = 1,
\end{aligned}
$$

Betweenness Centrality: Example



Node	Betweenness Centrality	Rank
A	16 + 1/2 + 1/2	1
B	7+5/2	3
C	0	7
D	5/2	5
E	1/2 + 1/2	6
F	15 + 2	1
G	0	7
H	0	7
I	7	4

Centrality in terms of how fast you can reach others

Closeness Centrality

Linton Freeman

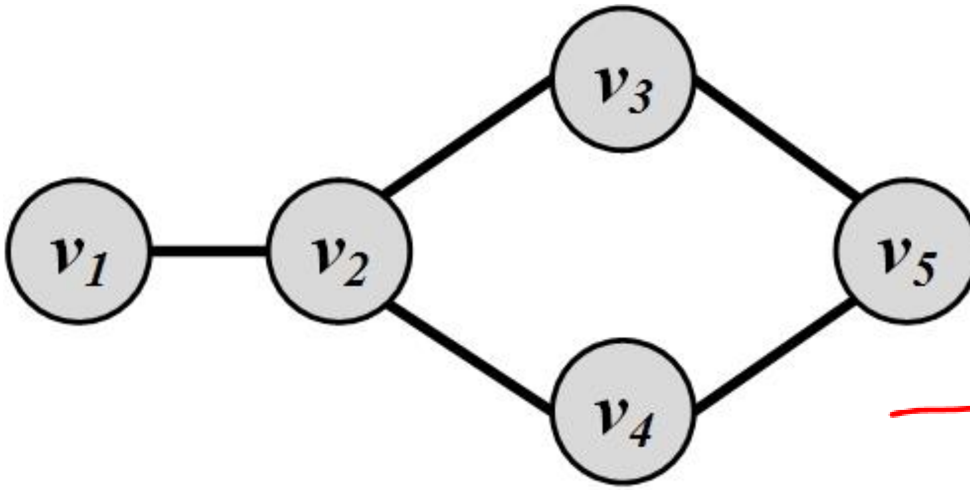
- The intuition is that influential/central nodes can quickly reach other nodes

- These nodes should have a smaller average shortest path length to others

Closeness centrality: $$C_c(v_i) = \frac{1}{\bar{l}_{v_i}}$$

$$\bar{l}_{v_i} = \frac{1}{n-1} \sum_{v_j \neq v_i} l_{i,j}$$

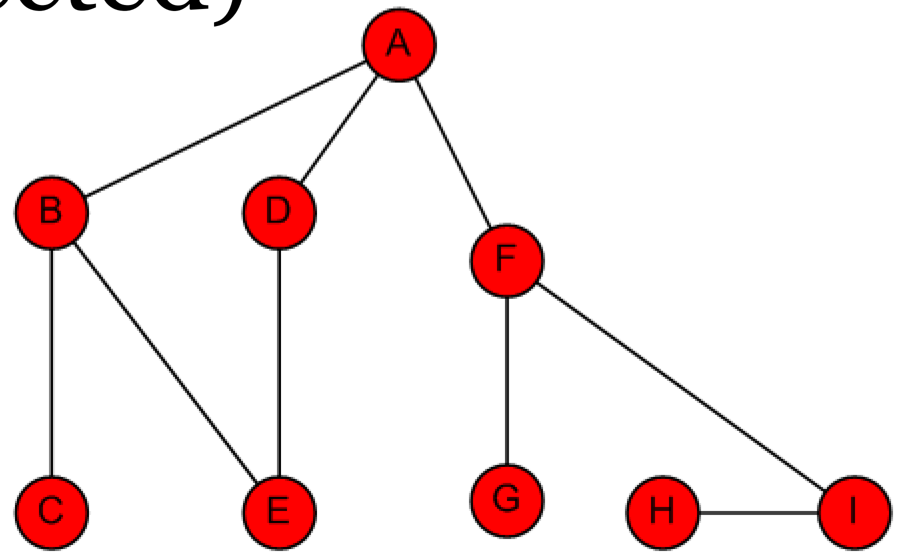
Closeness Centrality: Example 1



$$\frac{1}{\bar{l}_{v_i}} \qquad \frac{1+2+2+3}{x}$$

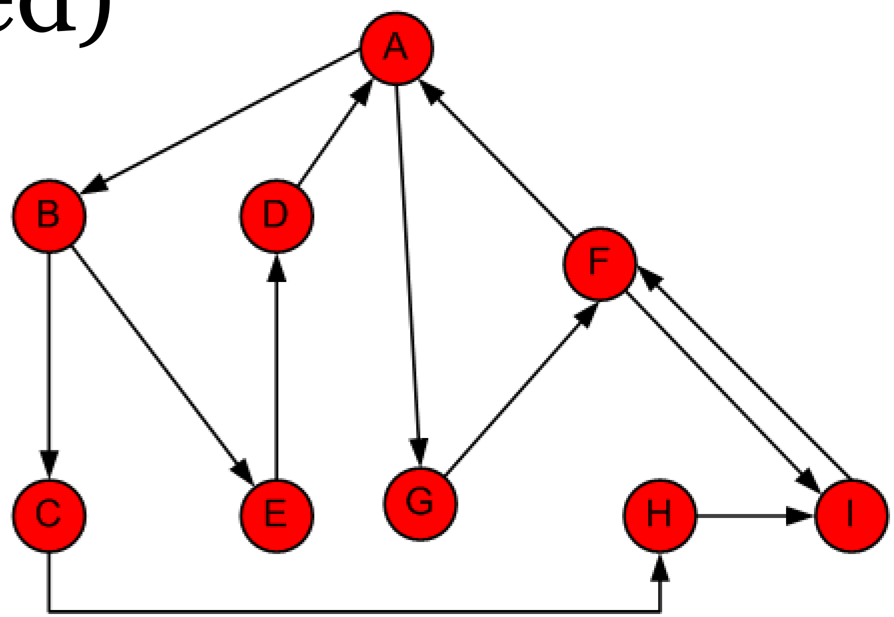
$$\begin{aligned} C_c(v_1) &= 1 \;/\; (\ (1+2+2+3)/4\ ) = 0.5, \\ C_c(v_2) &= 1 \;/\; (\ (1+1+1+2)/4\ ) = 0.8, \\ C_c(v_3) = \quad C_b(v_4) &= 1 \;/\; (\ (1+1+2+2)/4\ ) = 0.66, \\ C_c(v_5) &= 1 \;/\; (\ (1+1+2+3)/4\ ) = 0.57. \end{aligned}$$

Closeness Centrality: Example 2 (Undirected)



Node	A	B	C	D	E	F	G	H	I	D_Avg	Closeness Centrality	Rank
A	0	1	2	1	2	1	2	3	2	1.750	0.571	1
B	1	0	1	2	1	2	3	4	3	2.125	0.471	3
C	2	1	0	3	2	3	4	5	4	3.000	0.333	8
D	1	2	3	0	1	2	3	4	3	2.375	0.421	4
E	2	1	2	1	0	3	4	5	4	2.750	0.364	7
F	1	2	3	2	3	0	1	2	1	1.875	0.533	2
G	2	3	4	3	4	1	0	3	2	2.750	0.364	7
H	3	4	5	4	5	2	3	0	1	3.375	0.296	9
I	2	3	4	3	4	1	2	1	0	2.500	0.400	5

Closeness Centrality: Example 3 (Directed)



Node	A	B	C	D	E	F	G	H	I	D_Avg	Closeness Centrality	Rank
A	0	1	2	3	2	2	1	3	3	2.125	0.471	1
B	3	0	1	2	1	4	4	2	3	2.500	0.400	2
C	4	5	0	7	6	3	5	1	2	4.125	0.242	9
D	1	2	3	0	3	3	2	4	5	2.875	0.348	3
E	2	3	4	1	0	4	3	5	5	3.375	0.296	6
F	1	2	3	4	3	0	2	4	4	2.875	0.348	4
G	2	3	4	5	4	1	0	5	2	3.250	0.308	5
H	4	4	5	6	5	2	4	0	1	3.875	0.258	8
I	2	3	4	5	4	1	4	5	0	3.500	0.286	7

Questions?